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Outcomes of newly diagnosed idiopathic generalized epilepsy syndromes in a non-pediatric setting

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Introduction – The prognosis of idiopathic generalized epilepsy syndromes (IGES) in the adult setting may vary from that in children owing to differences in genetic, environmental and lifestyle factors. *Methods* – All patients diagnosed with epilepsy at the Epilepsy Unit, Western Infirmary, Glasgow, between 1981 and 2001 were reviewed. *Results* – Of 890 patients, 118 (13%) met the criteria for IGES. Outcomes were known for 103, 66 (64%) of whom achieved remission. The responder rate with sodium valproate was superior (66% vs 45%, $P = 0.073$) to that with lamotrigine (LTG) particularly in patients with juvenile myoclonic epilepsies (75% vs 39%, $P = 0.014$). History of febrile seizures was the only factor associated with reduced likelihood of remission ($P = 0.032$) *Conclusions* – Idiopathic generalized epilepsy syndromes constituted 13% of cases in a largely adult cohort of newly diagnosed epilepsy, most of whom achieved remission usually with a single antiepileptic drug. History of febrile seizures was associated with a poorer outcome.

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The International League Against Epilepsy (ILAE) Commission on Classification and Terminology defined idiopathic generalized epilepsy syndromes (IGES) as 'forms of generalized epilepsies in which all seizures are initially generalized [absences, myoclonic jerks and generalized tonic–clonic seizures (GTCS)], with an electroencephalographic (EEG) expression that is a generalized bilateral, synchronous, symmetrical discharge' (1) The interictal EEGs show normal background activity and generalized discharges, such as spikes, polyspike, spike-wave and 3 Hz polyspike-waves. Intellectual and neurological deficits are absent and no structural abnormalities can be demonstrated by brain imaging. The etiology is believed to be a genetic predisposition modified by environmental influences.

Idiopathic generalized epilepsy syndromes are believed to have a better prognosis than localization-related epilepsies (2, 3). Response to treatment, however, is governed by a number of factors, of which classification of the epilepsy is only one (4). Different lifestyle and environmental factors operate in adult patients compared to children, and treatment outcomes might not be translatable from studies in younger cohorts. An analysis of patients attending a regional epilepsy clinic in the UK found 54% response rate (1 year seizure freedom) overall for patients with IGES (5). This was a cross-sectional study in a clinic population and could, therefore, be expected to have an overrepresentation of patients with drug-resistant epilepsy. Studies recruiting patients from the time of diagnosis and incorporating a diagnostic scheme designed to identify IGES are required to assess accurately the prognosis of various syndromes. We examined the response to treatment in an unselected series of adolescent and adult patients with IGES diagnosed and treated at a single centre over a 20-year period.

Methods

Methods have been described in detail elsewhere (6). Adolescent and adult patients with suspected seizure disorders were referred to a specialist first seizure service by general practitioners, accident and emergency physicians and other clinicians. A structured protocol was used to collect clinical information and detailed histories were obtained from patients and witnesses. This analysis was planned when the study was initially set up in 1982. Seizure types and epilepsy syndromes were classified according to ILAE guidelines at the time of analysis (1, 7). The primary end point was seizure freedom for a minimum of 12 months on an unchanged treatment schedule. Those patients remaining seizure-free to the end of follow-up were regarded as being in remission. Patients who continued to report seizures were, by definition, considered to have uncontrolled epilepsy. Those whose seizures were initially controlled for a year or more but who subsequently became pharmacoresistant were categorized as having relapsed. The extent of control of seizures was assessed at the time of the patient's last hospital visit.

A total of 890 patients were diagnosed with epilepsy after presenting to the First Seizure Clinic at the Epilepsy Unit in the Western Infirmary in Glasgow, between August 1981 and May 2001. Sufficient follow-up data were available for 780 patients. Patients who did not return for review or were persistently non-adherent to their prescribed medication were excluded from analysis. Logistic regression analysis was used to ascertain the influence of various risk factors on prognosis for remission. Categorical variables were compared using the chi-squared test on 1 degree of freedom. Analyses were carried out using Minitab for windows statistical software (version 13.21).

Results

Patient demographics

Of the entire cohort of 890 adult and adolescent patients, 118 (13%) fitted the criteria for a diagnosis of IGES. Fifteen patients were lost to follow-up and the remaining 103 (47 males, 56 females) were included in the outcome analysis. The median age at epilepsy onset was 17 years (range 5–51) and that at diagnosis was 18 years (range 9–71). The age of onset for the various syndromes is shown in Fig. 1. Juvenile myoclonic epilepsy (JME) was the most common IGES syndrome and was present in 55 (53%) of cases, followed by epilepsy with GTCS only in 28 (27%) patients. The other syndromes identified were epilepsy with GTCS on awakening ($n = 7$), juvenile absence epilepsy ($n = 4$), epilepsy with GTCS during sleep ($n = 3$), late-onset myoclonic epilepsy ($n = 3$), childhood absence epilepsy ($n = 2$) and epilepsy with myoclonic absences ($n = 1$).

Response

Overall, 76 (74%) patients responded to treatment, 66 (64%) of whom achieved remission and 10 (10%) relapsed (Table 1). The remaining 27 (26%) patients did not attain seizure control for any

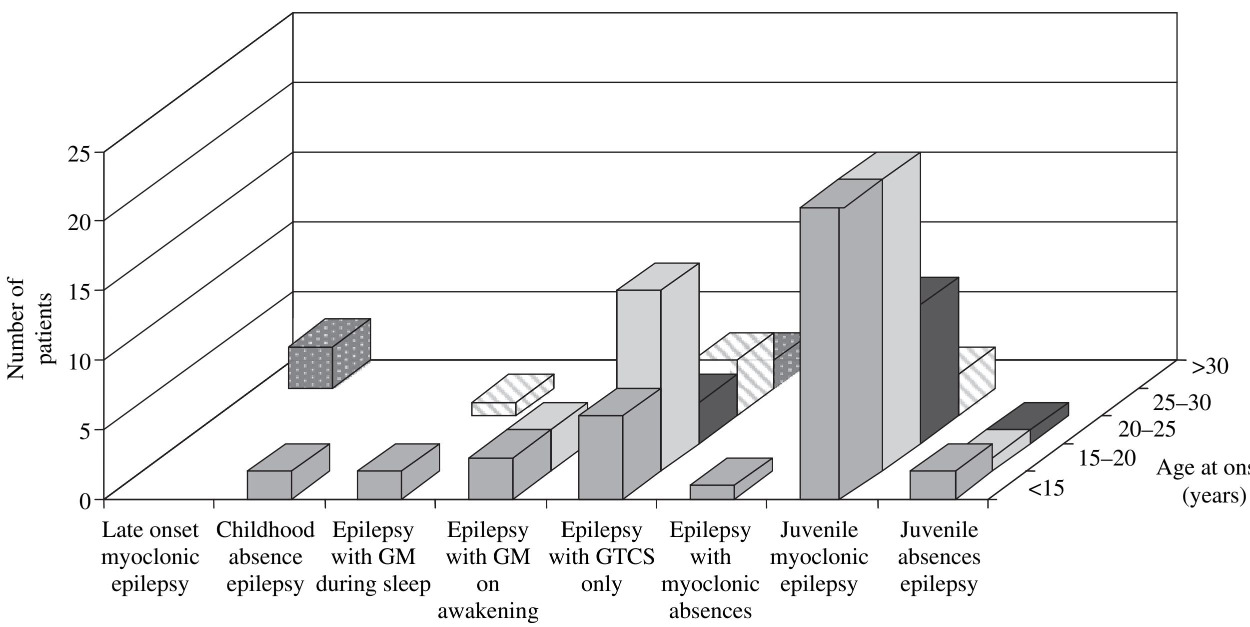

Figure 1. Age of seizure onset in patients diagnosed with idiopathic generalised epilepsy syndromes. GM, grand mal; GTCS, generalized tonic–clonic seizures.

Table 1 Treatment outcomes in 103 patients with newly diagnosed individual idiopathic generalized epilepsy syndromes

Syndrome	Remission (%)	Relapse (%)	Uncontrolled (%)	Total
Juvenile myoclonic epilepsy	40 (73)	4 (7)	11 (20)	55
Epilepsy with generalized tonic–clonic seizures	15 (54)	5 (18)	8 (29)	28
Epilepsy with generalized tonic–clonic seizures on awakening	5 (71)		2 (29)	7
Juvenile absence epilepsy	3 (75)		1 (25)	4
Late-onset myoclonic epilepsy		1 (33)	2 (67)	3
Epilepsy with generalized tonic–clonic seizures during sleep	2 (67)		1 (33)	3
Childhood absence epilepsy	1 (50)		1 (50)	2
Epilepsy with myoclonic absences			1 (100)	1
Total	66 (64)	10 (10)	27 (26)	103

12-month period. Treatment response was achieved with one antiepileptic drug (AED) (monotherapy) in 64 cases, 53 of whom (51%) responded to their first ever AED. When used as the first AED, valproate (VPA, 66%) was more likely to produce a good response than LTG (45%, $P = 0.073$). When response to the first AED in individual syndromes was analyzed, treatment with VPA was significantly more likely to be successful than that with LTG (75% vs 39%, $P = 0.014$) in patients with JME. Numbers of patients with other IGES syndromes were too small to make meaningful comparisons.

Remission

Of the 66 (64%) patients achieving remission, 33 (32%) had no further seizures after starting AED treatment and so were classified as immediate responders (6). The prognosis for immediate response was not influenced by the syndrome classification or the choice of first AED. Treatment outcomes in the individual IGES are shown in Table 1. Patients with JME achieved remission in 40 (73%) cases. The remission rate for JME was significantly better than in the rest of the cohort ($P = 0.05$). The number of patients with other subtypes of IGES was insufficient to demonstrate significant differences in outcomes among the various syndromes. Relapse occurred in four (7%) patients with JME and five (18%) patients with IGE with GTCS only.

Treatment leading to remission

Of the 66 patients achieving remission, 56 (85%) did so on AED monotherapy. VPA ($n = 30$) and lamotrigine (LTG, $n = 17$) were the most common AED producing remission, and were generally used

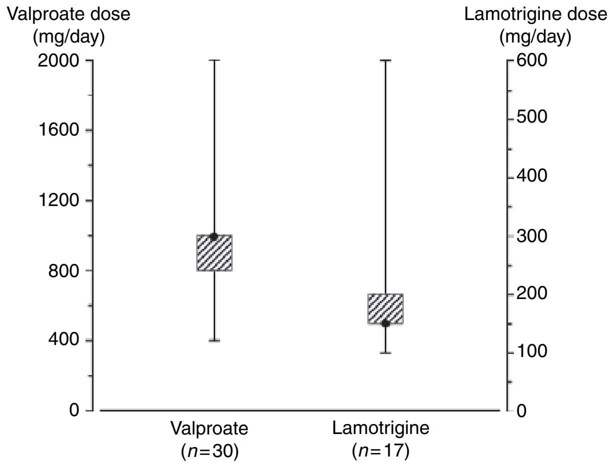

Figure 2. Median daily dose of valproate and lamotrigine producing remission in patients with newly diagnosed idiopathic generalized epilepsy syndromes. The error bars and hatched boxes represent the complete and interquartile dosage ranges.

in modest doses (Fig. 2). Of the patients failing initial monotherapy, 36 were tried with a second AED schedule with 22 receiving alternative monotherapy and 14 duotherapy. Seizure freedom was achieved by nine patients in each group (41% of those on alternative monotherapy and 64% of those on duotherapy). Overall, 47 patients entered remission on their first AED, 14 on their second regimen, four on their third and one on the fifth AED schedule. Duotherapy led to remission in 10 patients, eight of whom were receiving a combination of VPA with LTG, while one patient took VPA or LTG with topiramate.

Factors affecting prognosis

We explored the influence of potential risk factors, including age at onset, gender, family history of epilepsy, febrile seizures, psychiatric comorbidity, epileptiform abnormalities on EEG and types of seizures, on the likelihood of remission (Table 2). In a univariate logistic regression analysis, a history of febrile seizures was the only factor associated with a significantly lower remission rate (odds ratio 0.18, 95% confidence intervals 0.03–0.96, $P = 0.032$).

Discussion

Idiopathic generalized epilepsy syndromes constitute 20–40% of all new diagnoses of epilepsy (8, 9). Although seizures generally begin in childhood and adolescence, onset after the second decade of life has long been described (10). Panayiotopoulos et al. reported a syndrome of adult-onset IGES

Table 2 Univariate analysis of factors that may affect the prognosis of idiopathic generalized epilepsy syndromes

Risk factor	Patients	Remission	%	Odds ratio	95% CI	*P*
Gender						
Female	47	30	64	0.98	0.44–2.20	0.962
Male	56	36	64			
Family history						
Present	14	7	50	0.46	0.15–1.43	0.183
Absent	86	59	69			
Febrile seizures						
Present	7	2	29	0.18	0.03–0.96	0.032
Absent	92	64	70			
Psychiatric comorbidity						
Present	9	6	67	1.13	0.27–4.82	0.865
Absent	94	60	64			
EEG						
Epileptiform	87	55	63	0.78	0.25–2.45	0.669
Normal/non-specific	16	11	69			
Seizure types						
Tonic-clonic only	37	20	54	1.45	0.64–3.30	0.373
Other seizure combinations	66	42	64			
Age of onset						
<15	37	25	68	1.27	0.54–2.97	0.579
15–20	38	23	61	0.78	0.34–1.80	0.567
20–25	15	12	80	2.52	0.66–9.58	0.148
25–30	8	4	50	0.53	0.12–2.27	0.396
>30	5	2	40	0.35	0.06–2.22	0.262

with phantom absences (3–4 s lapses in consciousness), infrequent generalized GTCS and frequent absence status in 13 of a series of 86 patients with adult-onset IGES (11). Gilliam et al. described 11 patients with onset of myoclonic epilepsy at a mean age of 39 years, some of whom also had absences (12). A case series from a regional epilepsy clinic in the UK identified 8.5% patients with onset of IGES after the age of 20 years, based on clinical features and seizure types (13). A similar series from New York, based on retrospective review of clinical and EEG data, identified 13.3% patients with adult-onset IGES (14). A rigorous approach to classification could identify many more such patients, as demonstrated by Marini et al. They employed post-ictal EEG within 24 h of the initial seizure followed a sleep-deprived investigation if necessary and identified adult-onset IGES in 28% of cases (15).

Recognition of IGES in the non-pediatric setting has important implications for management. Early diagnosis of IGES can provide reassurance for adult patients and unnecessary brain imaging can be avoided. Identification is also important for clinical genetic studies and regulatory trials of new AEDs. However, the greatest value of accurate classification is in guiding the choice of treatment. Narrow-spectrum AEDs, such as phenytoin and carbamazepine, can exacerbate seizures in patients with IGES (16, 17). There have also been reports of worsening of myoclonus by LTG (18, 19). Benbadis et al. reported that 48% of patients with IGES seen in a specialist epilepsy program had previously been started on inappropriate AEDs (20). In our cohort, the highest response rate to initial monotherapy was observed with VPA, traditionally regarded as the drug of choice for IGES (21). This was especially so in patients with JME, where treatment with VPA was clearly superior to that with LTG. There is now substantial potential to improve the outcome in JME and other IGES with some of the modern broad-spectrum AEDs such as topiramate, levetiracetam, and zonisamide (22–24).

From their study of 962 pediatric and adult patients with IGES, Nicolson et al. found atypical presentation to be associated with a significantly worse response to treatment (5). This was defined as IGES with onset outside the expected age group of 3–20 years or with atypical absence or myoclonic seizures. Patients with GTCS only (64.5% remission) or GTCS on awakening (78% remission) did significantly better than the rest of the cohort. Fernando-Dongas et al. reported a higher prevalence of EEG asymmetry, atypical seizure characteristics and intellectual deficiency in patients with JME who failed to respond to treatment with VPA (25). Psychiatric problems have also been reported to be associated with drug-resistant IGES (14, 26). In our cohort, a history of febrile seizures was the only factor associated with a failure to achieve remission. Febrile seizures are known to have a genetic component and some of the genetic factors associated with IGES can give rise to febrile seizures (27). It is possible that the IGES associated with febrile seizures have a different pathophysiology and, therefore, require a different therapeutic strategy (28).

Conclusions

The majority of patients with IGES presenting at an adult epilepsy clinic had an excellent prognosis usually with a single AED. VPA was more efficacious than LTG, especially in patients with JME. Previous febrile seizures were associated with poorer outcome. Prospective studies recruiting patients from the time of diagnosis and employing diagnostic schemes designed to identify specific IGES will help to elucidate their natural history.

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